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Lesson 5: Experimental Methodology

Technologies for Artificial Intelligence

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Before we start: lesson 4, concluded

- First, lesson 4’s last slides: receiver operating characteristic (ROC) curves, the area under the curve (AUC), the threshold from costs, notebook 3
- Exercise 4 is set at their end; exercises 4 and 5 are both discussed on 13 November
- Then lesson 5, with a few parts left for self-study, marked (optional)

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We broke a promise on purpose

- Lesson 4 chose a threshold **by looking at the test set**
- Lesson 1 forbade exactly that
- Today we pay the debt

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Today: how to tell whether it works

- How far apart can two honest splits be?
- Cross-validation: what it estimates, and what it does not
- Bias, variance, and the error no model can remove
- Learning curves: more data, or a different model?
- Leakage that survives cross-validation
- Lesson 4’s threshold, done properly

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A test score is a measurement

- You would not report a poll of forty people to three decimals
- Holding data out makes the estimate **unbiased**: right on average over all the test sets you might have drawn, for the model you fitted
- It does nothing to make it **precise**: close to the truth on the one split you did draw

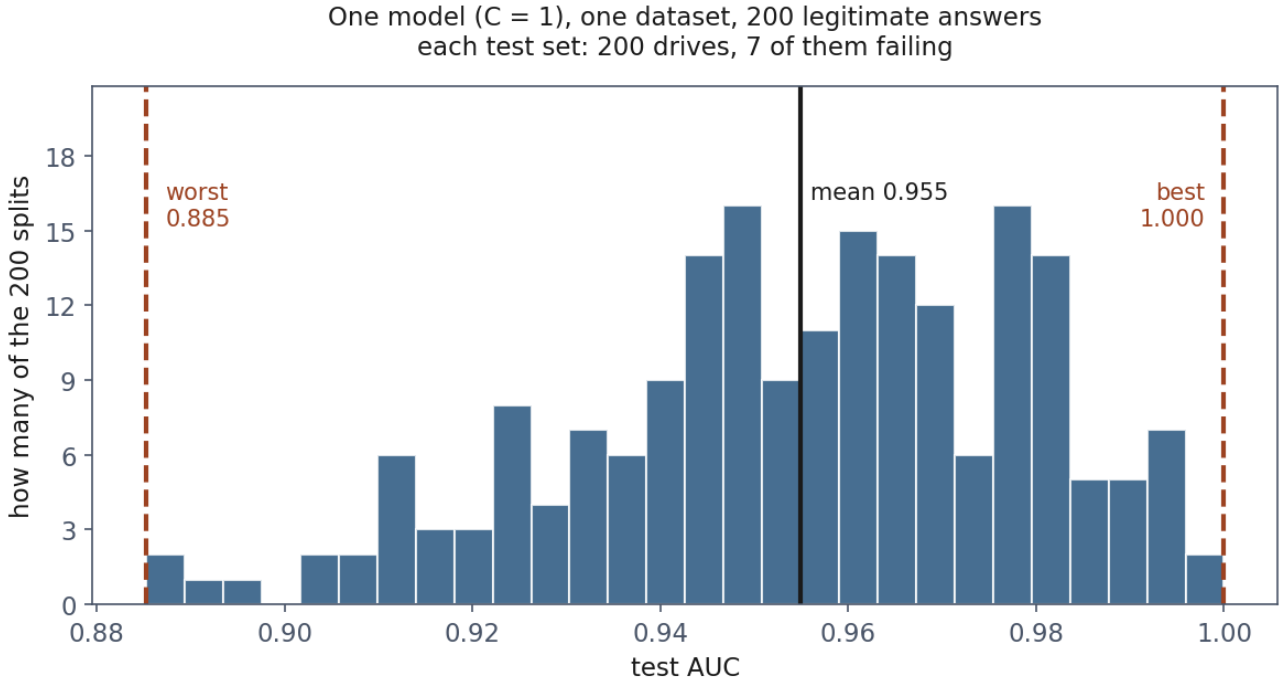
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The fleet, cut down

- Lesson 4’s drives, cut to **800**, of which **29 fail** (3.6%)
- Why 800: real projects often have a few hundred rows, a few dozen positive
- In code: `train_test_split(X, y, test_size=0.25, stratify=y)`; lesson 4’s pipeline fitted on the training part; `roc_auc_score` on the test part

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200 honest splits, one dataset



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Why do the 200 scores spread so widely?

- Each test set takes 25% of the 29 failures: 0.25 × 29 = 7.25, so **7 failures** among 200 drives
- AUC = pairs ranked right ÷ 1,351, the 7 × 193 = 1,351 pairs of one failing and one healthy drive
- For example, one failing drive slipping below 20 more healthy ones costs 20 ÷ 1,351 = 0.015 of AUC

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Parameters and hyperparameters

- **Parameters**: estimated by the fit, w and b
- **Hyperparameters**: not estimated by the fit; some fixed before it (penalty λ, neighbours, tree depth), some after it (the threshold)
- Training error would always pick $\lambda = 0$, so they are chosen by **comparing** models on data the fit did not see
- scikit-learn's C plays the role of $1/\lambda$: smaller C, stronger penalty

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Choosing a hyperparameter: two recipes

Fixed before the fit: the penalty (C), tree depth, neighbours

1. Set the test set aside

↓

2. For each candidate, C = 0.01, 1, 100:
5-fold cross-validation on the rest
= 5 fits each, 3 × 5 = 15 fits

↓

3. Keep the C with the best mean score

↓

4. Refit with that C on all non-test data
(fit number 16)

↓

5. Score the test set, once

Chosen after the fit: the decision threshold

1. Set the test set aside,
then cut a validation set from the rest

↓

2. Fit the model on the training part:
one fit in all

↓

3. Predict probabilities on the validation set

↓

4. Try every threshold on those probabilities;
keep the cheapest

↓

5. Score the test set, once

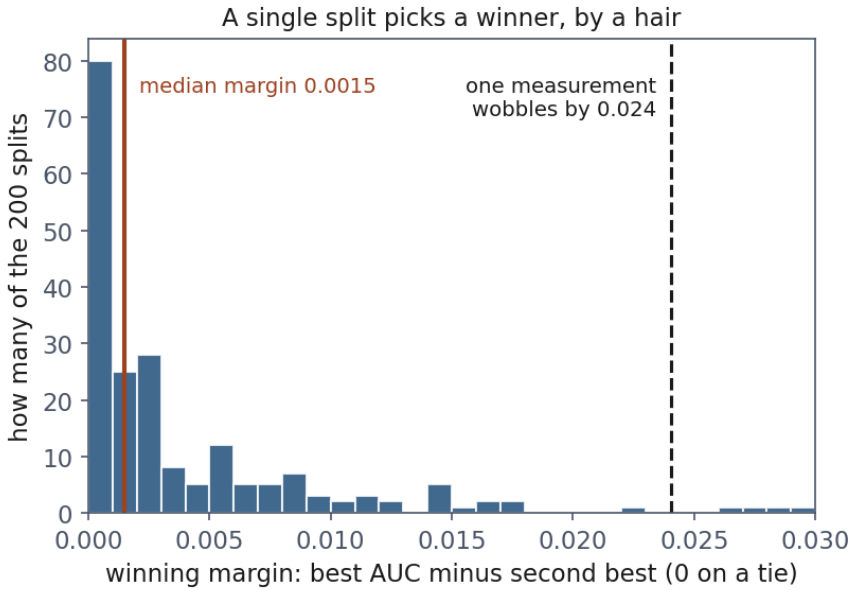
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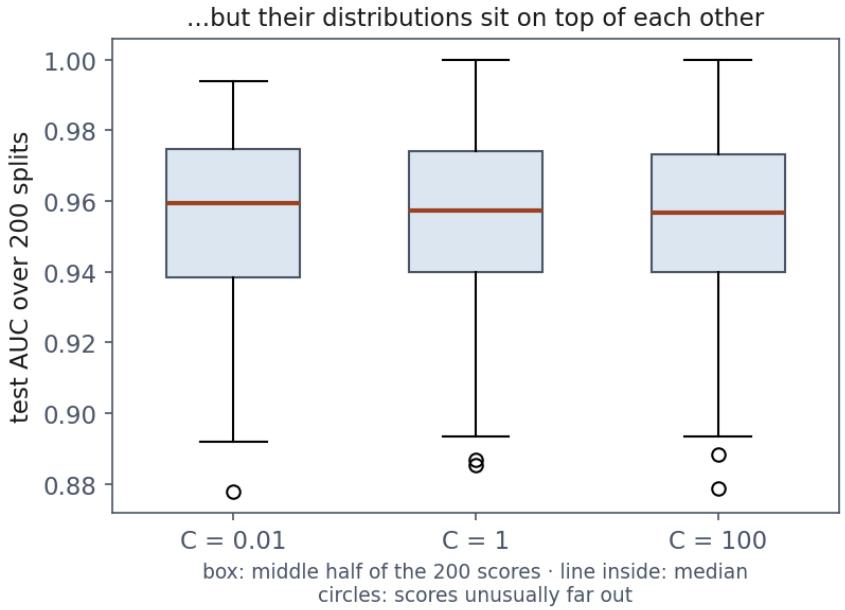
On each of slide 7’s 200 splits, three models compete

Model (C = 1, the default, is slide 7’s)	Mean test AUC over the 200 splits	Splits it won outright (31 were ties)
C = 0.01	0.9549	89
C = 1	0.9549	42
C = 100	0.9544	38

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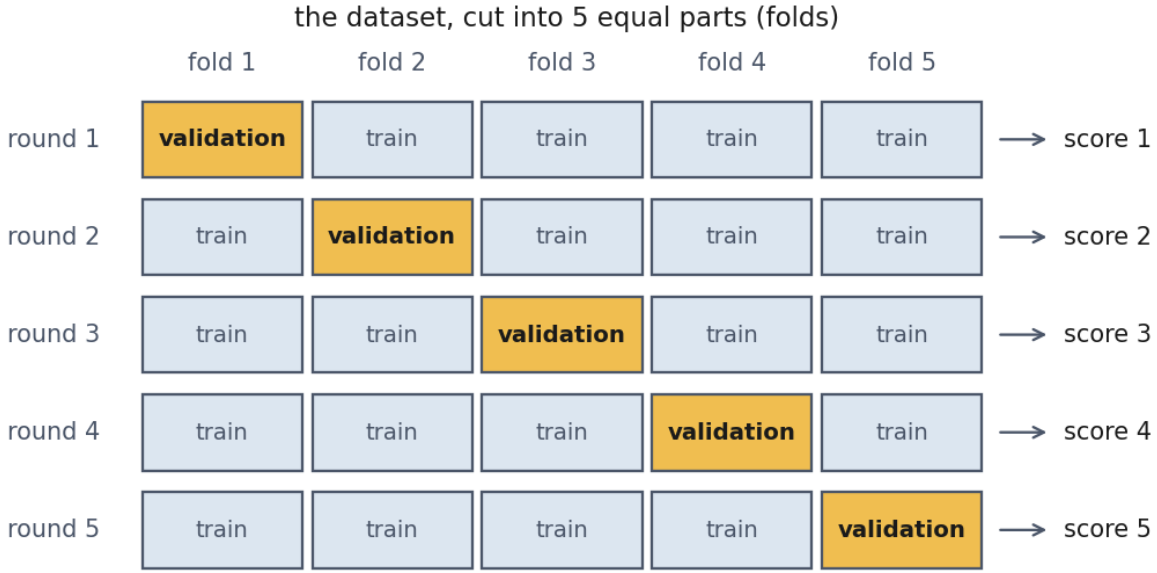
The same three models: who wins, and how they score





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k-fold cross-validation



estimate =
(score 1 + ... + score 5) ÷ 5

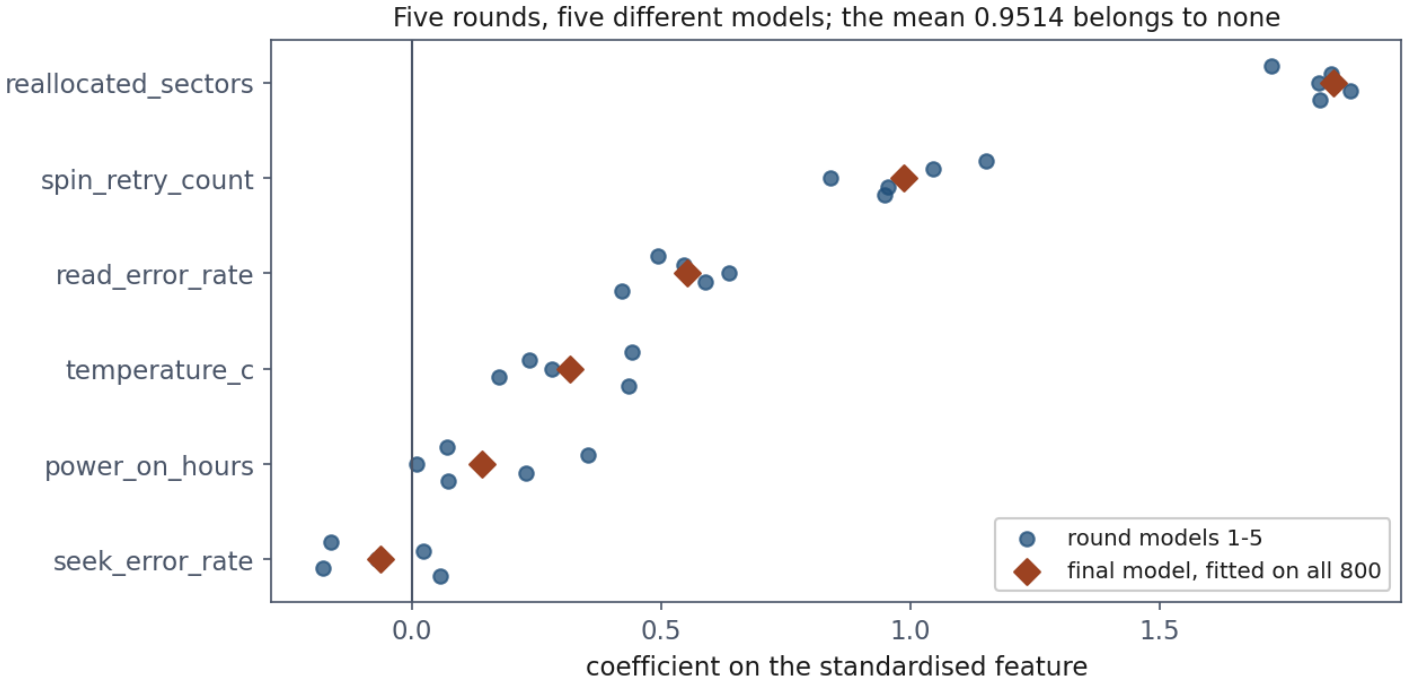
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What it actually estimates

- The performance of a **procedure** (the whole recipe), not of one fitted model
- Usually slightly **pessimistic**: each model trains on 4/5 = 80% of the data
- Seed 0: mean of the five round AUCs, **0.9514**
- In code: `cross_val_score(model, X, y, cv=StratifiedKFold(5, shuffle=True, random_state=0), scoring="roc_auc")`

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Five rounds, five different models



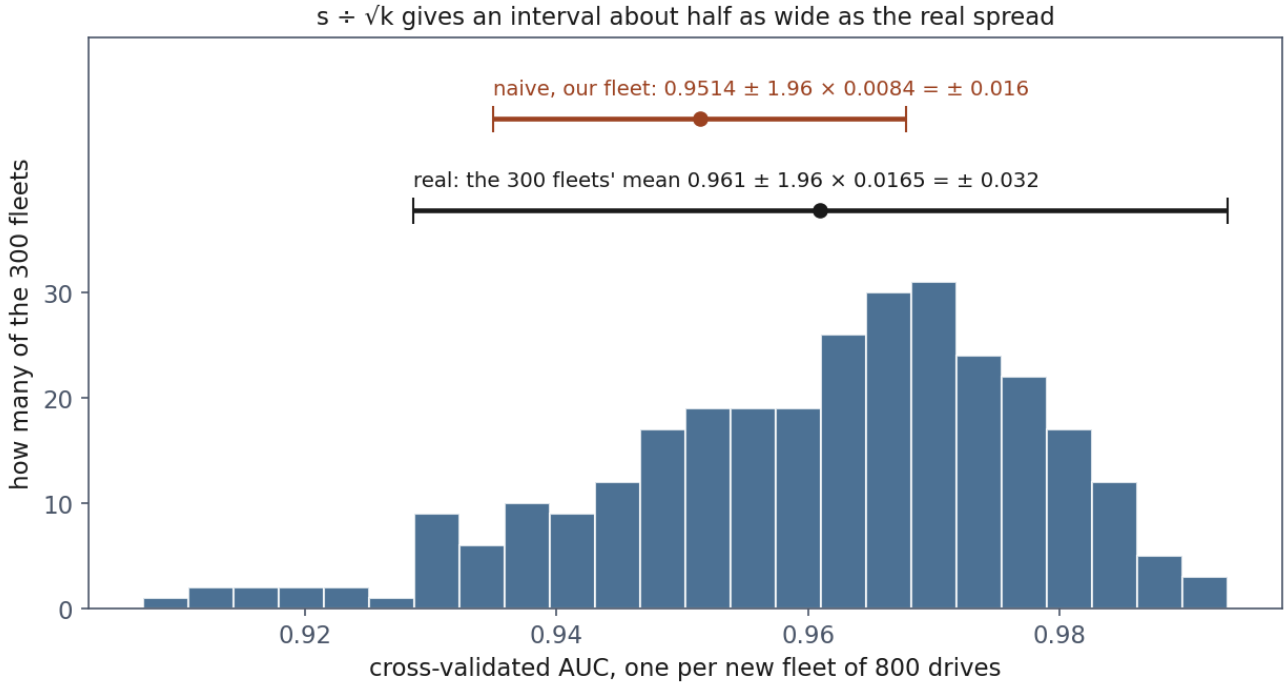
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Report the spread, but not as a confidence interval

- Approximate 95% **confidence interval** (normal approximation): estimate ± 1.96 × standard error; it would contain the truth in about 95 of 100 repeated experiments
- s ÷ √k (s: standard deviation of the k round scores) is **too small**: training sets share 3 of 4 folds, and s rests on only 5 scores
- Report the mean and s, call s a spread, treat small differences as unresolved

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On 300 new fleets, the real spread is twice the naive one

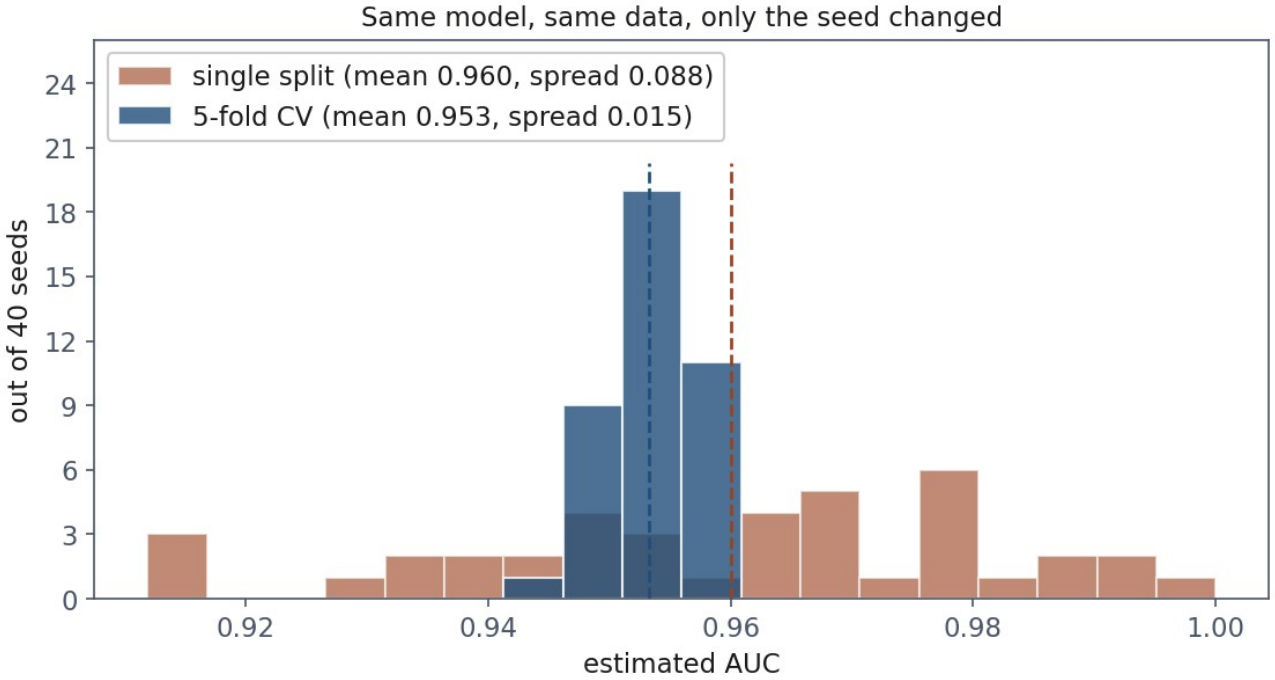


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Stratify when a class is rare

Fold (seed 3)	KFold: folds cut at random	StratifiedKFold: each fold keeps 3.6% failures
1	7 failures of 160, AUC 0.979	5 of 160, AUC 0.983
2	**1 failure of 160, AUC 1.000**	6 of 160, AUC 0.969
3	7 of 160, AUC 0.942	6 of 160, AUC 0.902
4	5 of 160, AUC 0.965	6 of 160, AUC 0.945
5	9 of 160, AUC 0.943	6 of 160, AUC 0.969

Change only the seed, 40 times: how far does the score move? (optional)



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How many folds? (optional)

- **k = 5 or 10** in practice
- Larger k: less pessimistic, more fits to run, round scores can be more correlated
- k = m (m the number of rows) is **leave-one-out**: every row is a validation fold on its own

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When the rows are not independent

- Suppose the table held ten readings per drive (lesson 4’s has one row per drive)
- A random 5-fold split puts about eight of a drive’s ten readings in training and two in validation, in every round
- The model recognises the drive, not the failure

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Split by the thing you must generalise to

- If one drive (or patient, or customer) has several rows: **GroupKFold** keeps all its rows in the same fold
- The final test split too: no drive on both sides (`GroupShuffleSplit`)
- If the model must predict the future: **TimeSeriesSplit** trains on the past and validates on what comes after it

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Notebook 1, live

section, and its slides	you call (→ the number to come back with)
two honest experiments, 5-6	`train_test_split(..., random_state=seed)`, `make_pipeline(StandardScaler(), LogisticRegression())`, `roc_auc_score` → seed 3 gives 1.000
how wide, 7-8	the same, 200 times → 0.885 to 1.000
choosing on one split, 9-12	three values of C, a winner per split → 89, 42, 38 outright wins, 31 ties
k-fold by hand, 13-17	`StratifiedKFold`, then `cross_val_score` → mean 0.9514; 300 new fleets → 0.0165 against 0.0084
stratify, 18-19	`KFold` against `StratifiedKFold` → 1 failure, AUC 1.000; 0.0881 against 0.0149

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Break

- Ten minutes

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Error too high: more data, or another model?

- Daily energy use from the outdoor temperature, as in lesson 3
- A straight line on 25 days spread over the year: validation mean squared error (MSE) **6,064.3 kWh²**
- **More days** or **another model**: each cures a different error, the wrong one nothing
- To choose, find where the error comes from: first where we know the truth, then on real data

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A lab where we know the truth

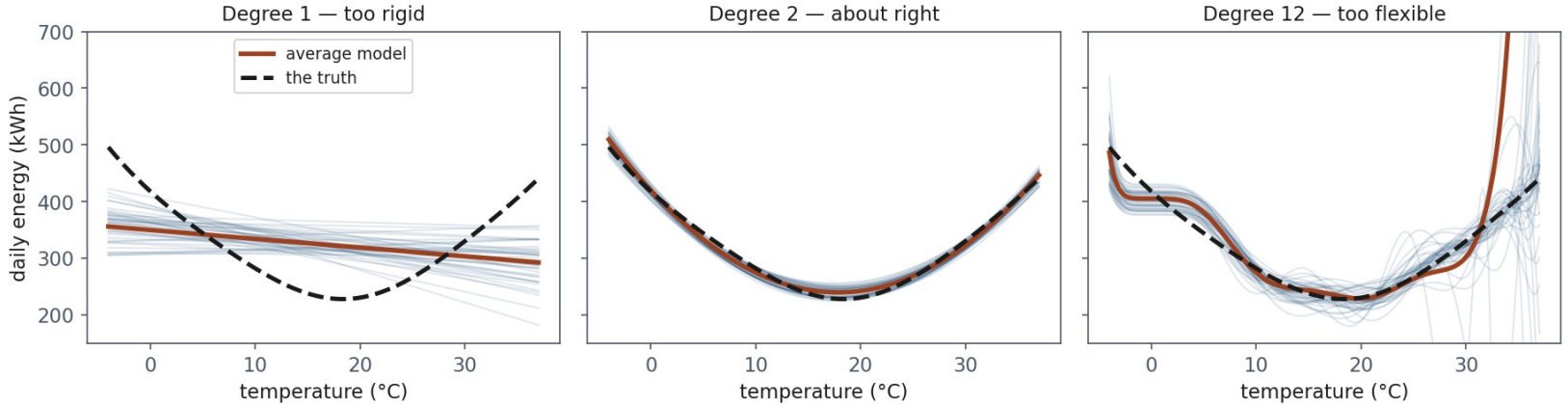
- Truth: $f(t) = 240 + 0.55\,(t - 18)^2 + 12\sin(t/4)$ kWh, t = daily mean temperature (°C); noise 22 kWh (standard deviation)
- 300 new training sets of 25 days at −5 to 38 °C, each drawn afresh (slide 25’s is the first)
- Polynomials of degree 1 to 12:

```
make_pipeline(PolynomialFeatures(d),
LinearRegression())
```


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Three hundred parallel universes: one model each

Each faint line is one model, fitted to its own 25 days (40 of the 300 drawn). Bias: the red average against the black truth. Variance: how widely the faint lines spread around the red average.



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Expected error on a new day = bias² + variance + noise

$$\mathrm{E}\left[(y - \hat{f}(x))^2\right] = \mathrm{bias}^2 + \mathrm{variance} + \sigma^2$$

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What each term means

- A new day at x = 30 °C: y = its energy (kWh); $\hat{f}(x)$ = ONE model's prediction; $\mathbb{E}$ = average over models and noise
- **Bias** = average prediction − truth (kWh, signed); the mean squared error, expanded, yields bias²
- **Variance** = average of (prediction − average prediction)²: the models' disagreement
- **Noise** $\sigma^2 = 22^2 = 484$: the days' scatter; known because we added it

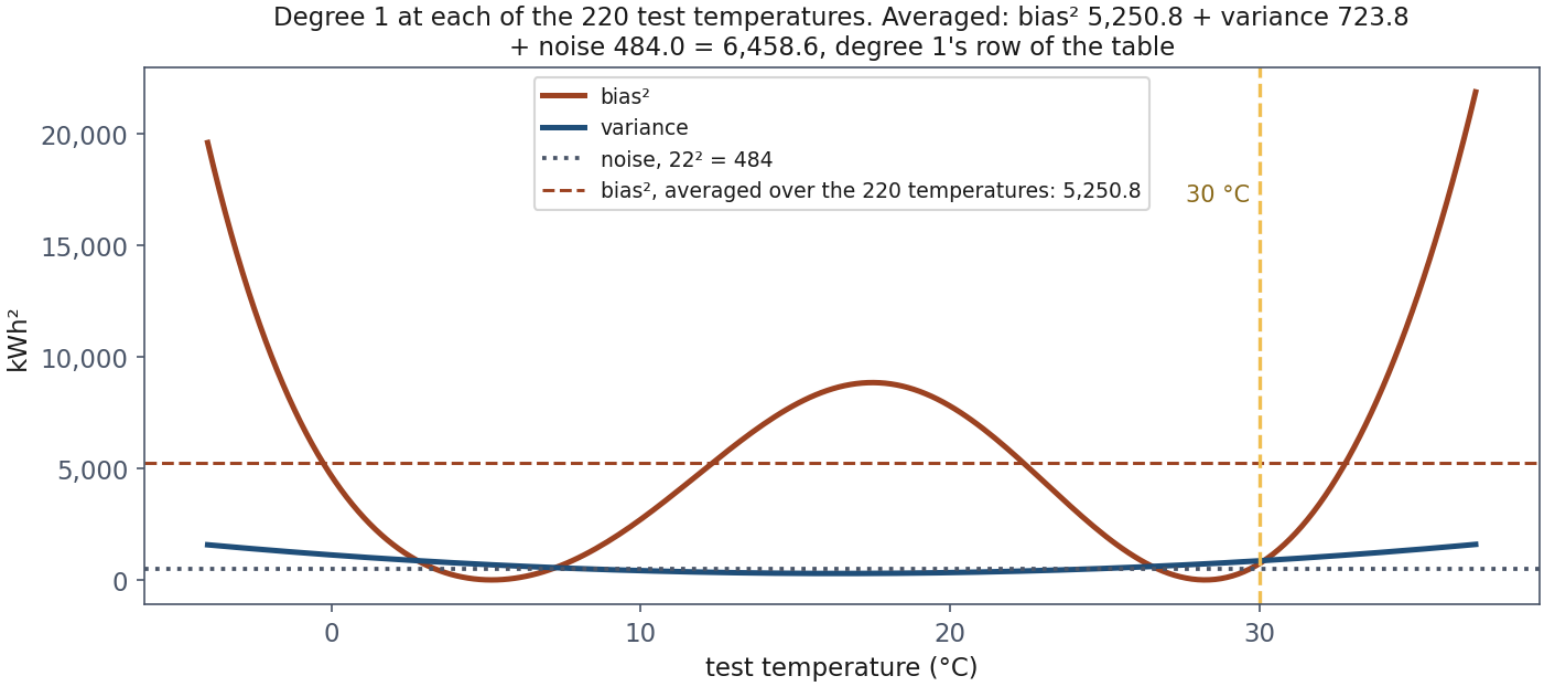
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The straight line (degree 1) at 30 °C, in three numbers

Step, degree 1	Formula	Value
The truth	f(30), known because we wrote f	330.5
The 300 predictions	one per model	300.8, 317.1, 304.8, and so on, from 215.6 to 364.7
Their average	sum ÷ 300	303.6
bias²	$(303.6 - 330.5)^2$	**724**
Variance	sum of $(\text{prediction} - 303.6)^2$ ÷ 300	**856**
Noise	22^2	**484**
Expected error	bias^2 + variance + noise	**2,064**

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Degree 1 at 220 temperatures, −4 to 37 °C, then averaged (optional)



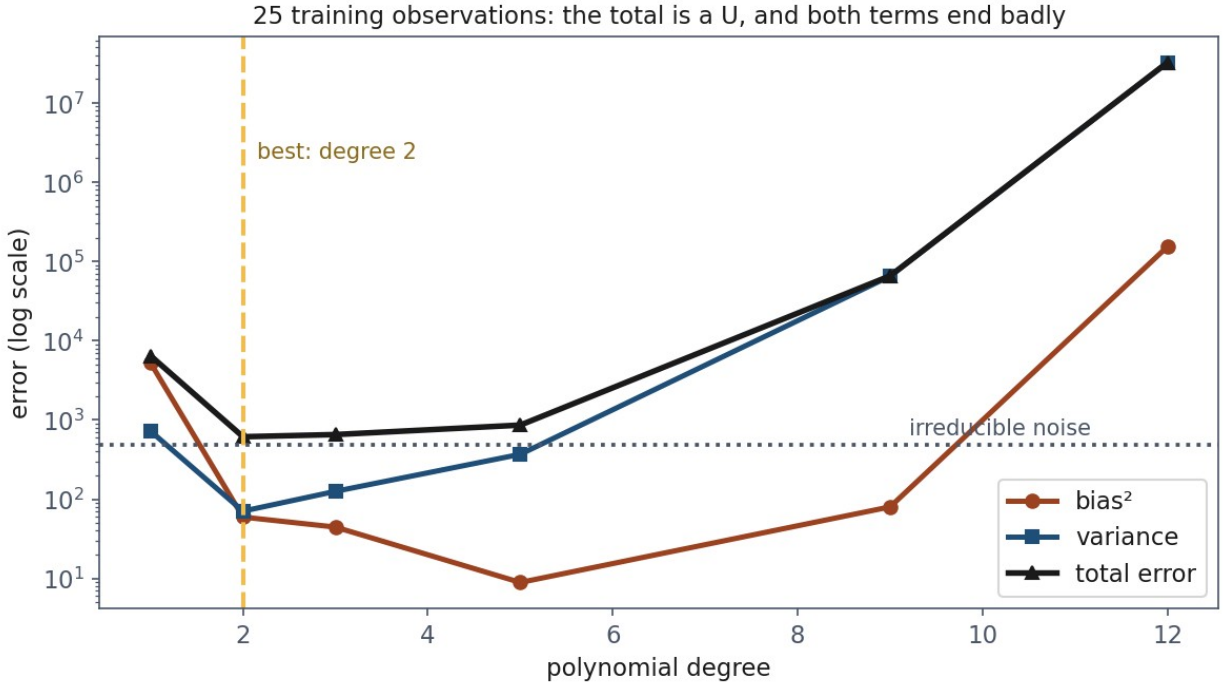
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Every degree, on 25 days (optional)

Degree	bias²	Variance	Noise	Total = bias² + variance + noise
1	5,250.8	723.8	484.0	6,458.6
2	59.6	70.8	484.0	**614.3**
3	44.5	125.8	484.0	654.4
5	8.9	368.9	484.0	861.8
9	80.1	65,186.0	484.0	65,750.1
12	153,537.3	32,174,344.9	484.0	32,328,366.2

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The total error is a U: too rigid on the left, too flexible on the right



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The floor you cannot go below (optional)

- Noise variance is **484** = 22^2, the noise put into the data, at every complexity
- A **test** error well below a known floor means leakage or a mistake in the evaluation, not excellence

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The best model depends on how much data you have (optional)

Degree; total error at training-set size m (bold: lowest in its column)	m = 25	m = 60	m = 150
1	6,458.6	5,998.6	5,861.5
2	**614.3**	570.6	555.3
5	861.8	**540.9**	**506.9**
12	32,328,366.2	7,626.9	899.0

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Without the truth: two errors you can still measure

- One training set, f and σ unknown: cross-validation inside it, test set untouched, gives two errors
- **Training error**, on the days the model was fitted to: it has followed them, so it hides its variance
- **Validation error**, on the fold it did not see: it pays bias, variance and noise
- Training error too high for the task: **bias**. Validation far above training: **variance**

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One sample of 25 days: the two errors tell what the lab told

Degree	Training MSE, kWh²	Validation MSE, kWh²	Reading	What the lab measured
1	5,155.7	6,064.3	both high and close: **bias**	bias² 5,250.8
2	312.8	**450.4**	lowest validation: **the best**	lowest total, 614.3
5	262.4	615.1	training below the noise, 484; a gap: **variance**	variance 368.9 against degree 2's 70.8

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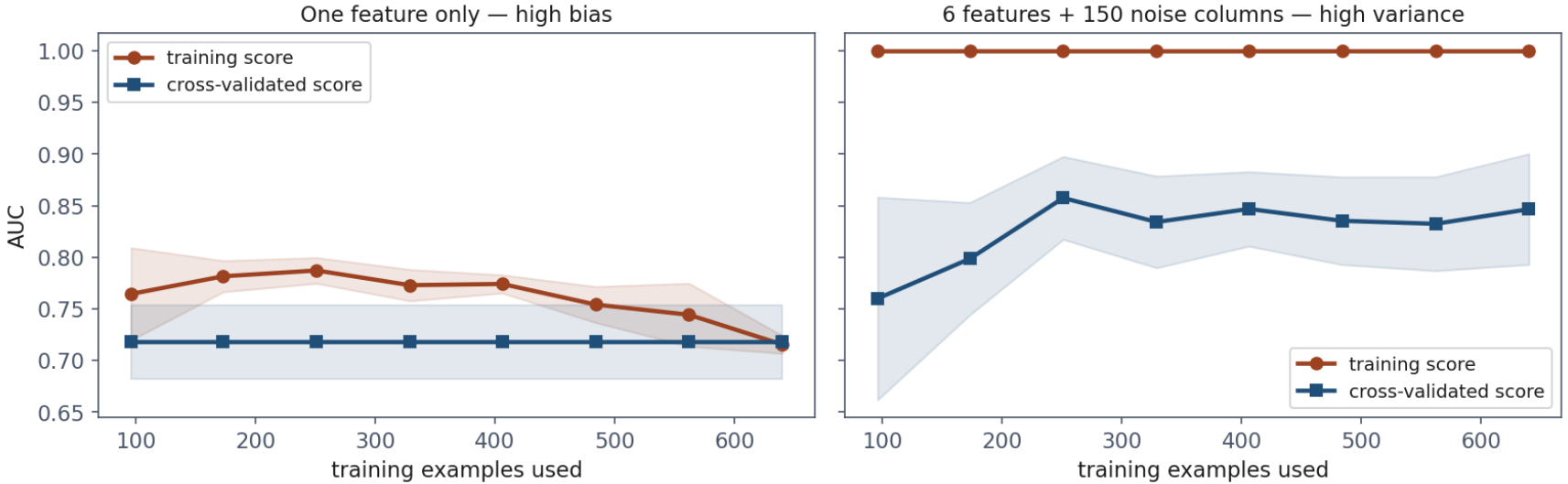
Learning curves: would more data help?

- Slide 37’s two errors, repeated at growing training-set sizes: `learning_curve(model, X, y, cv=folds, scoring="roc_auc")`
- On lesson 4’s 800 drives, scored by the area under the curve (AUC): **higher is better**, so poor = **low** curves
- Two models broken on purpose: one feature only; 6 features + 150 noise columns, almost no penalty

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Learning curves: left the curves meet low, right a gap that stops closing

Lesson 4's 800 drives, logistic regression, 5-fold cross-validation. The score is the AUC: higher is better, so a poor model shows as LOW curves. Bias: the two curves meet low. Variance: a gap between them.



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Which fix to buy

- Curves meet at a **poor** score: bias. More data will **not** help; change the model
- Wide gap, validation **still rising** by more than its spread: more data **likely** helps
- Wide gap, validation **flat**: stronger penalty (smaller C) or **remove features** (slide 39, right)
- Slide 25’s line is bias: degree 2, same 25 days, **450.4**
- The test set: scored once, after the fix

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Notebook 2, self-study (optional)

section, and its slides	you call (→ the number to come back with)
300 parallel universes, 26-27	`PolynomialFeatures` then `LinearRegression`, in `make_pipeline`, fitted 300 times
the three parts measured, 28-34	NumPy on the 300 predictions → sum and measured error agree to 2.73e-12
more data moves the answer, 35	the same at 25, 60 and 150 days → best degree 2, then 5
without the truth, 25 and 36-37	`cross_validate` with `KFold`, `return_train_score=True` → degree 1: 5,155.7 against 6,064.3
learning curves, 38-40	`learning_curve` with `StratifiedKFold` → 0.715 against 0.718; 1.000 against 0.847

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Now: what cross-validation does not protect you from

- Labels: slide 6’s 800 drives, 29 failures; features: **2,000 columns of random noise**
- No link to the label: honest AUC **0.500** for any model
- Keep the 10 columns with the highest F-test (between-class variance ÷ within-class variance), `SelectKBest(f_classif, k=10)`; then cross-validate a logistic regression on them: 5 rounds, train on 640 drives, score 160

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The code almost everyone writes

```
model = make_pipeline(StandardScaler(),
LogisticRegression(max_iter=2000))
cv = StratifiedKFold(5, shuffle=True,
random_state=0)
sel = SelectKBest(f_classif, k=10)
sel.fit(noise, y)
X = noise.loc[:, sel.get_support()]
cross_val_score(model, X, y, cv=cv,
scoring="roc_auc")
```


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0.931

- Cross-validated AUC **0.931** on data with no signal; folds 0.928, 0.979, 0.975, 0.943, 0.832
- The bug is `sel.fit(noise, y)`: the selector saw **every row**, validation folds included
- It kept the 10 columns that best matched the labels it would later be graded on

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Fix it, and look what happens

- Selection moved **inside** the pipeline, refitted in every round: `Pipeline([("select", SelectKBest(f_classif, k=10)), ("scale", StandardScaler()), ("model", LogisticRegression())])`
- 20 different fold seeds, 0 to 19: **0.658** on average, 0.576 to 0.759
- The truth is still 0.500

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Why it does not return to 0.5

- With 2,000 columns and 800 rows, some columns match the label **by accident**
- That accident belongs to the **sample**, not to the split
- Every fold contains it
- What helps: a test set set aside **before** any search, fewer candidates, or more rows

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GridSearchCV: try all, keep the best

- **k** = columns kept: 2, 5, 10, 20, 40 (**columns, not folds**)
- **C** = penalty, smaller is stronger: 0.01, 0.1, 1, 10, 100; **25 pairs**
- Each pair: 5-fold cross-validation, 25 × 5 = **125 fits**
- Best mean AUC wins: `best_params_` (k = 20, C = 10); `best_score_` = that mean
- Then refitted on all 800, **not scored again**

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The best of 25 scores is inflated: 0.800 for a worthless model

`GridSearchCV`: k = 2, 5, 10, 20, 40 columns × C = 0.01, 0.1, 1, 10, 100	AUC
best of the 25 candidates, `best_score_`	**0.800**
average candidate	0.726
worst candidate	0.684
nested cross-validation, next slide	0.670
the truth, by construction	0.500
optimism of the search, 0.800 − 0.670	0.130

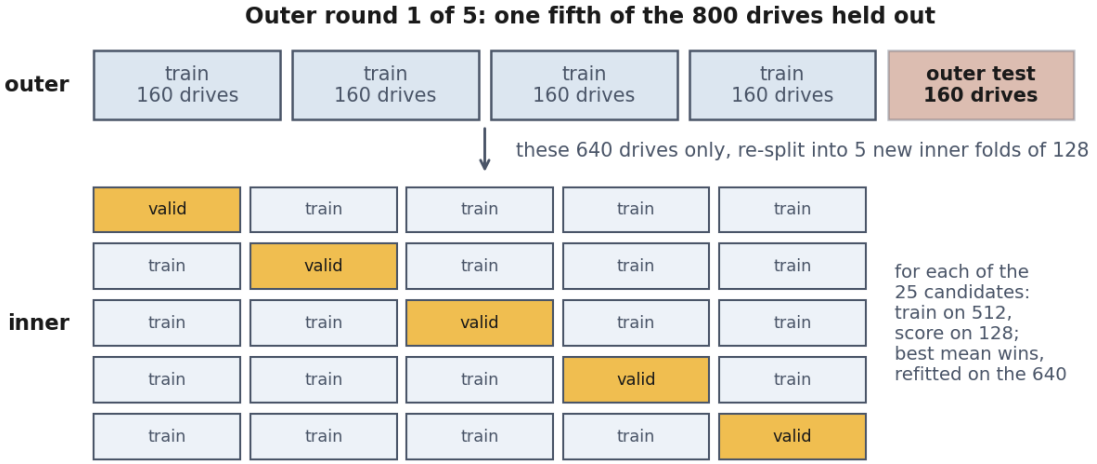
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Cross-validate the whole search

- Ordinary cross-validation scores **one fixed** model; `GridSearchCV`'s folds **choose and score**: `best_score_` is the luckiest of 25
- Nested:
  `cross_val_score(GridSearchCV(honest, grid, cv=inner), noise, y, cv=outer)`
- Each outer round: the search on 640 drives picks a winner, scored **once** on the other 160
- Mean of the 5 outer scores: **0.670**

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Nested cross-validation: it measures what choosing costs



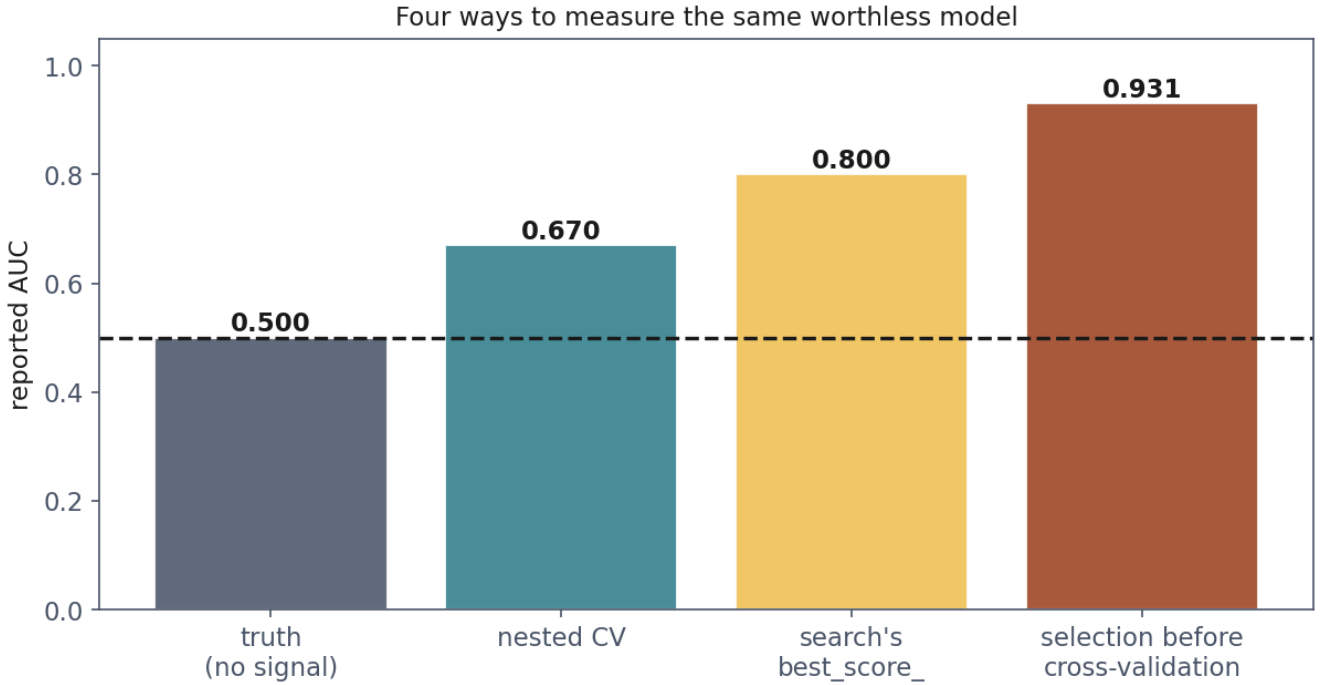
The winner is scored once on the 160 outer test drives: 0.698.

5 outer rounds: (0.698 + 0.852 + 0.754 + 0.487 + 0.558) ÷ 5 = 0.670

It measures what choosing costs; it does not choose.

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Each piece of discipline costs score: 0.931, 0.800, 0.670, 0.500



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The debt, paid: report 4,700 EUR, the threshold chosen on validation

Threshold, for one logistic regression fit on 450 of slide 6's 800 drives	missed failures × 2,600 EUR + false alarms × 140 EUR	Cost on the 200 test drives (7 failures)
0.50, the default: nothing chosen	4 × 2,600 + 1 × 140	10,540 EUR
0.050: the cheapest of 200 thresholds from 0.01 to 0.9, tried on the 150 **validation** drives	1 × 2,600 + 15 × 140	**4,700 EUR**
0.171: the same sweep, on the 200 **test** drives (lesson 4's way)	1 × 2,600 + 5 × 140	3,300 EUR

Three sets, three jobs

- **Train**: fit the parameters
- **Validation**: make every choice: model, hyperparameters, threshold
- **Test**: report. Touched **once**
- Change anything after looking at the test set, and it has become a validation set

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A fixed seed buys repeatability, not stability (optional)

- 30 seeds of the train/test split (600 drives to fit, 200 to test), same model: **0.913 to 1.000**
- Every run is perfectly reproducible
- `AUC 1.000 (random_state=3)` is reproducible **and** misleading

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What to fix, and what to write down (optional)

- Fix every `random_state` (the split, the folds), and seed your own NumPy draws with `np.random.default_rng(seed)`: a rerun gives the same number
- **Library versions**: defaults change between releases
- What you could **not** fix: parallel computations that finish in a different order, graphics processing unit (GPU) non-determinism

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Notebook 3, live

section, and its slides	you call (→ the number to come back with)
the catastrophe, 42-44	`SelectKBest(f_classif, k=10)` on every row, then `cross_val_score` → 0.931
selection inside the round, 45-46	`Pipeline` with the selector as its first step → 0.658
the search, 47-48	`GridSearchCV`, `best_score_` → 0.800
nested, 49-51	`cross_val_score(GridSearchCV(honest, grid, cv=inner), noise, y, cv=outer)` → 0.670
the threshold, honestly, 52-53	`train_test_split` twice → 4,700 EUR against 10,540
seeds, 54-55	30 values of `random_state` → 0.913 to 1.000

What to put in a report

- The metric, its **spread** (standard deviation of the fold scores, slide 16), and how it was estimated (e.g. stratified 5-fold, seed 0)
- How many **positives** the test set held
- Every seed, and every library version
- Which choices were made on which data

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What to take away

- **A score is a measurement**: 0.885 to **1.000**
- **Choosing on one split chooses noise**
- **Training error shows bias, the gap variance**
- **Cross-validation is honest only about what runs inside it**. It cannot remove
  - chance correlations in the sample: 0.658 on pure noise
  - the luck of choosing: 0.800 reported, 0.670 nested
  - dependent rows, unless split with `GroupKFold`

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Homework

- **Exercise 5**, discussed at the start of **Friday 13 November**
- A result that is too good. Find out why, then produce the honest number

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